

WALVIS BAY, Namibia

WMO: 680980

Lat: 22.980S

Lon: 14.645E

Elev: 299

StdP: 14.54

Time Zone: 2.00 (E02)

Period: 04-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	42.4	44.3	15.6	12.2	70.0	21.2	16.1	73.0	27.8	77.1	23.1	70.9	5.4	60	0.434

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	18.4	95.0	64.6	91.3	62.3	87.4	62.4	70.3	84.9	68.5	81.7	67.1	79.1	14.8	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
65.8	96.3	76.4	64.2	90.8	73.9	62.7	86.0	71.7	34.5	85.3	32.9	82.0	31.7	79.3	77.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%		2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
Min		Max		Min	Max		Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)		(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.0	19.5	16.5		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	65.0	69.0	69.3	69.8	68.6	65.6	63.5	63.3	58.1	59.9	62.0	64.1	66.6
	DBStd	6.69	2.91	3.38	5.38	7.34	7.28	8.98	8.72	4.81	3.39	3.17	3.10	2.71
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	968	2	2	6	32	78	141	148	231	160	103	51	14
	CDD50	5460	589	541	613	559	483	405	413	250	296	373	423	515
	CDD65	954	127	122	154	140	96	97	96	16	7	11	24	64
	CDH74	7460	373	475	1124	1399	1046	1039	1099	253	123	124	198	207
	CDH80	2976	45	85	463	642	476	495	513	119	40	33	49	16
(14) Wind	WSAvg	7.3	8.0	7.5	7.0	6.6	6.6	7.7	7.7	6.9	7.2	7.6	7.6	7.6
(15) Precipitation	PrecAvg	1.3	0.2	0.2	0.4	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.1
	PrecMax	3.1	1.3	1.6	1.9	1.0	0.2	0.1	0.3	0.1	0.4	0.2	0.5	0.6
	PrecMin	0.3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	0.9	0.3	0.4	0.4	0.3	0.0	0.0	0.1	0.0	0.1	0.0	0.1	0.2
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	84.4	88.9	99.7	98.7	97.0	96.8	93.6	91.7	84.7	88.6	86.1	82.6
		MCWB	68.1	69.7	70.4	67.5	63.7	61.3	60.4	59.3	58.0	64.6	66.1	64.5
	2%	DB	80.5	82.4	91.8	92.9	91.3	91.6	91.0	81.7	75.6	77.9	78.4	79.0
		MCWB	66.4	67.3	69.6	66.4	61.8	58.9	59.5	56.1	59.0	61.1	61.7	63.9
	5%	DB	78.0	79.1	84.6	87.1	84.6	85.9	86.3	73.2	71.4	73.6	75.3	76.7
		MCWB	65.6	66.2	66.9	65.9	61.4	58.2	58.6	54.6	57.6	59.6	60.9	63.5
	10%	DB	75.5	77.0	79.5	81.8	78.8	79.2	80.4	68.2	68.2	70.9	72.8	74.3
		MCWB	64.8	65.5	66.1	64.2	59.5	56.0	57.0	55.6	57.5	59.0	60.3	62.8
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	68.9	70.6	74.6	72.0	68.5	63.1	65.7	60.7	61.9	66.1	66.0	66.9
		MCDB	80.6	84.6	88.7	85.6	83.3	88.1	79.1	79.9	76.1	84.8	77.0	78.1
	2%	WB	67.5	68.6	70.9	69.4	64.3	60.9	63.2	58.4	60.2	62.9	63.1	65.2
		MCDB	77.3	79.7	85.6	86.2	82.1	83.3	77.5	73.4	72.3	75.3	74.4	76.2
	5%	WB	66.4	67.3	68.6	67.2	62.4	59.4	60.9	57.1	58.9	60.8	61.8	64.1
		MCDB	75.9	77.3	82.5	83.7	81.0	79.0	77.0	68.8	69.2	71.7	72.6	74.4
	10%	WB	65.3	66.0	66.3	64.7	61.0	58.1	58.9	56.0	57.6	59.4	60.7	63.1
		MCDB	73.8	75.3	78.5	79.5	75.7	74.6	77.3	66.8	66.6	69.1	71.1	73.1
(35) Mean Daily Temperature Range	5% DB	MDBR	14.2	15.7	18.4	21.5	23.8	24.5	24.5	22.5	19.9	18.8	18.9	16.2
		MCDBR	17.4	18.5	25.3	27.5	30.6	29.5	31.7	32.2	27.0	24.3	23.2	19.6
		MCWBR	6.8	7.2	9.0	10.1	11.5	12.8	14.0	15.6	12.9	10.9	10.3	7.9
	5% WB	MCDBR	14.9	17.0	21.9	24.0	26.2	26.7	27.4	24.2	21.5	20.5	20.0	17.0
		MCWBR	6.2	6.8	8.8	9.9	11.3	13.0	13.3	12.8	10.6	9.7	9.4	7.3
(40) Clear-Sky Solar Irradiance	taub		0.436	0.443	0.416	0.371	0.340	0.328	0.348	0.393	0.456	0.454	0.400	0.408
	taud		2.254	2.257	2.314	2.402	2.424	2.414	2.362	2.233	2.025	2.082	2.284	2.279
	Ebn at Noon		289	283	282	281	278	275	272	270	265	276	298	298
	Edh at Noon		47	46	42	35	31	30	33	41	55	54	45	46
(44) All-Sky Solar Radiation	RadAvg		2397	2249	2011	1707	1426	1303	1361	1548	1812	2135	2435	2494
	RadStd		97	95	73	77	54	44	43	70	78	66	111	89

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page